

Fake News Detection Using Machine Learning

Likhila Vydana

Department of Electronics and Communication Engineering (Artificial Intelligence)

Indira Gandhi Delhi Technical University for Women (IGDTUW)

Delhi, India

Email: likhila.vydana@gmail.com

Abstract—The rapid growth of digital media platforms has significantly increased the spread of fake news, making it difficult for users to distinguish between authentic and misleading information. The widespread dissemination of false information can influence public opinion, create social unrest, and reduce trust in online news platforms. Therefore, developing an automated fake news detection system has become an important research problem.

This paper presents a machine learning-based approach for fake news detection using Natural Language Processing (NLP) techniques. The proposed system utilizes a publicly available Kaggle Fake News dataset containing real and fake news articles. The textual data undergoes preprocessing, including lowercasing, punctuation removal, stopword elimination, and lemmatization to improve text quality. Term Frequency-Inverse Document Frequency (TF-IDF) is employed to convert textual information into numerical feature vectors suitable for machine learning models.

Multiple classification algorithms, including Logistic Regression, Multinomial Naïve Bayes, Random Forest, and K-Nearest Neighbors, are evaluated for performance. Experimental results show that Random Forest achieved the highest classification accuracy of 99.84%, while Logistic Regression demonstrated competitive performance with 98.96% accuracy and was selected as the deployment model due to its lower computational complexity and faster inference. The trained Logistic Regression model was integrated into a Streamlit-based web application that enables users to classify news articles in real time. The proposed approach provides an efficient and reliable solution for automated fake news detection.

Index Terms—Fake News Detection, Machine Learning, Natural Language Processing, TF-IDF, Logistic Regression, Text Classification, Streamlit

I. INTRODUCTION

The internet has transformed the way people access and consume news. Social media platforms, online news portals, and digital communication channels allow information to spread globally within seconds. Although this rapid dissemination of information offers numerous benefits, it also facilitates the widespread circulation of fake news. Fake news refers to deliberately fabricated or misleading information presented as legitimate news with the intention of influencing public opinion, generating financial gains, or creating social and political instability.

The increasing volume of online content makes manual verification of news articles impractical. Consequently, automated fake news detection has emerged as an active area of research in machine learning and natural language processing. By analyzing linguistic patterns and textual characteristics,

machine learning algorithms can effectively distinguish between authentic and fabricated news articles.

Traditional fact-checking methods rely heavily on human experts and require considerable time and resources. In contrast, machine learning models can process thousands of articles within seconds while maintaining high classification accuracy. Recent advances in text preprocessing, feature extraction, and supervised learning algorithms have significantly improved the effectiveness of automated fake news detection systems.

In this work, a machine learning-based fake news detection system is proposed using TF-IDF feature extraction and multiple classification algorithms. Four machine learning models, namely Logistic Regression, Multinomial Naïve Bayes, Random Forest, and K-Nearest Neighbors, are trained and evaluated using a publicly available benchmark dataset. Experimental evaluation shows that Random Forest achieved the highest classification accuracy, while Logistic Regression was selected for deployment because of its computational efficiency, faster inference, and suitability for real-time prediction through a Streamlit web application.

The major objectives of this work are:

- To preprocess textual news data using Natural Language Processing techniques.
- To transform textual information into numerical representations using TF-IDF vectorization.
- To train and compare multiple supervised machine learning algorithms for fake news classification.
- To evaluate the models using standard performance metrics including Accuracy, Precision, Recall, and F1-score.
- To develop a user-friendly web application for real-time fake news prediction.

II. LITERATURE REVIEW

Recent advancements in machine learning and natural language processing have significantly improved the effectiveness of automated fake news detection systems. Researchers have explored various supervised learning algorithms and deep learning techniques to classify textual information based on linguistic patterns and semantic features.

Shu et al. proposed a comprehensive framework for fake news detection by combining social context, user behavior, and news content, demonstrating that textual analysis plays a crucial role in misinformation identification. Ahmed et al. evaluated traditional machine learning algorithms including

Logistic Regression, Naïve Bayes, and Support Vector Machines, reporting that TF-IDF combined with supervised learning provides high classification accuracy for textual datasets.

More recent studies have investigated transformer-based language models such as BERT and RoBERTa, which achieve state-of-the-art performance through contextual language understanding. Although these models offer superior accuracy, they require significantly higher computational resources and longer training times.

Considering computational efficiency, ease of deployment, and excellent predictive performance, traditional machine learning models remain suitable for real-time fake news detection applications. Therefore, this work evaluates multiple machine learning algorithms using TF-IDF feature extraction and selects the best-performing model for deployment.

III. DATASET DESCRIPTION

The dataset used in this study was obtained from the Kaggle Fake News Detection repository. It consists of two publicly available datasets containing genuine and fake news articles collected from multiple online news sources.

The fake news dataset contains 23,481 articles, whereas the true news dataset contains 21,417 articles, resulting in a combined dataset of 44,898 news articles. Each record includes the title, article text, subject category, publication date, and the corresponding class label.

During preprocessing, the title and article body were merged into a single textual feature named content. The target labels were assigned as 0 for fake news and 1 for genuine news. The dataset was randomly shuffled before dividing it into training and testing subsets to ensure unbiased model evaluation.

TABLE I
DATASET SUMMARY

Parameter	Value
Dataset Source	Kaggle Fake News Dataset
Fake News Articles	23,481
Real News Articles	21,417
Total Articles	44,898
Features	Title, Text, Subject, Date
Target Labels	Fake (0), Real (1)
Feature Extraction	TF-IDF

IV. PROPOSED METHODOLOGY

The proposed fake news detection system consists of four major stages: data preprocessing, feature extraction, model training, and prediction. Figure 1 illustrates the overall workflow of the proposed system.

Initially, the fake and genuine news datasets are merged into a single dataset. The title and article text are combined into a single feature named content to improve contextual understanding. The dataset is then preprocessed using Natural Language Processing (NLP) techniques before feature extraction.

The cleaned textual data is transformed into numerical vectors using the Term Frequency–Inverse Document Frequency

(TF-IDF) technique. These vectors are then used to train multiple supervised machine learning algorithms. Finally, the best-performing model is deployed through a Streamlit-based web application for real-time news classification.

A. Data Preprocessing

Raw textual data often contains punctuation, special symbols, stopwords, and inconsistent letter casing, which may negatively affect model performance. Therefore, several preprocessing operations were performed before feature extraction.

The preprocessing pipeline includes the following steps:

- Conversion of all text to lowercase.
- Removal of punctuation marks and numerical characters.
- Elimination of English stopwords using the Natural Language Toolkit (NLTK).
- Lemmatization to reduce words to their root forms.
- Removal of extra whitespace and unwanted symbols.

After preprocessing, the title and article body were merged into a single textual feature called content, which was used for feature extraction and model training.

B. Feature Extraction

Machine learning algorithms cannot directly process textual data. Therefore, the cleaned news articles were converted into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) vectorization technique.

TF-IDF assigns higher weights to words that frequently appear in a particular document but occur less frequently across the entire corpus. This enables the model to identify informative words while reducing the influence of commonly occurring terms.

The TF-IDF score is computed as follows:

$$TF-IDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right)$$

where $TF(t, d)$ represents the frequency of term t in document d , $DF(t)$ denotes the number of documents containing the term, and N represents the total number of documents in the corpus.

C. Machine Learning Models

Four supervised machine learning algorithms were trained and evaluated for fake news classification.

1. Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of a news article belonging to either the fake or genuine class. It provides fast training, efficient prediction, and excellent performance on high-dimensional sparse textual data.

2. Multinomial Naïve Bayes

Multinomial Naïve Bayes is a probabilistic classifier that assumes conditional independence among textual features. It performs well on text classification problems due to its computational efficiency.

3. Random Forest

Random Forest is an ensemble learning algorithm consisting of multiple decision trees. It improves classification performance by combining predictions from several trees while reducing overfitting.

4. K-Nearest Neighbors (KNN)

KNN is a non-parametric algorithm that classifies samples based on the labels of their nearest neighboring instances in the feature space. Although simple to implement, its computational complexity increases with larger datasets.

D. Model Training and Evaluation

The processed dataset was divided into training and testing subsets using an 80:20 ratio. TF-IDF vectorization was fitted on the training data and subsequently applied to the testing data to prevent data leakage.

Each machine learning model was trained using the same feature representation and evaluated using Accuracy, Precision, Recall, and F1-score. The model achieving the highest overall performance was selected for deployment in the Streamlit web application.

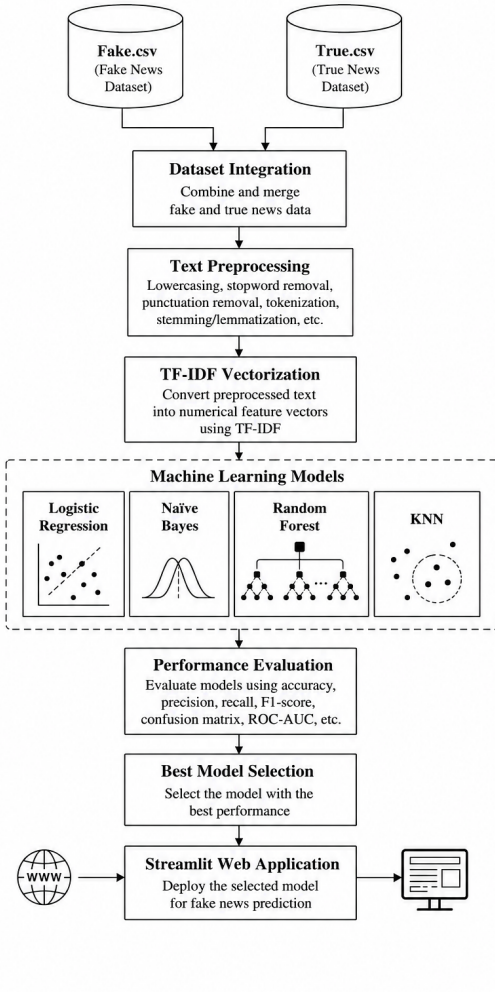


Fig. 1. Proposed methodology for fake news detection.

V. EXPERIMENTAL RESULTS

The proposed system was implemented in Python using Scikit-learn and NLTK and evaluated on the Kaggle Fake News dataset. The dataset was split into training and testing sets (80:20). Four supervised machine learning models were trained using TF-IDF features and evaluated using Accuracy, Precision, Recall, and F1-score.

A. Performance Metrics

The models were evaluated using Accuracy, Precision, Recall, and F1-score. Accuracy measures overall correctness, Precision indicates the proportion of correct positive predictions, Recall measures the detection of actual positives, and F1-score provides a balanced evaluation of Precision and Recall.

B. Model Comparison

TABLE II
MODEL COMPARISON

Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	98.96	0.99	0.99	0.99
Multinomial Naïve Bayes	94.77	0.94	0.95	0.95
Random Forest	99.84	1.00	1.00	1.00
K-Nearest Neighbors	72.45	0.94	0.45	0.61

Random Forest achieved the highest classification accuracy (99.84%), while Logistic Regression achieved a competitive accuracy of 98.96% and was selected for deployment due to its lower computational complexity and faster inference.

C. Confusion Matrix

The confusion matrix compares actual and predicted class labels, demonstrating the effectiveness of the proposed classifier.

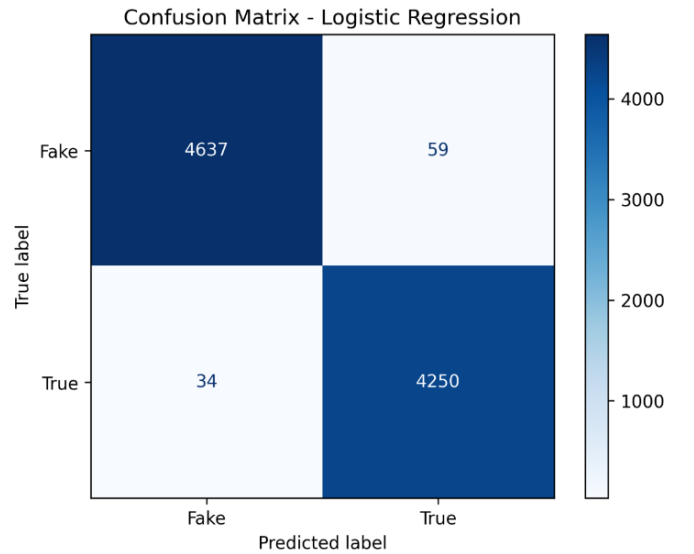


Fig. 2. Confusion Matrix of the Logistic Regression classifier.

D. Classification Performance

The results demonstrate that TF-IDF feature extraction combined with Logistic Regression provides highly discriminative textual representations for fake news detection. The preprocessing pipeline significantly reduces noise within the textual data, enabling the classifier to learn meaningful patterns from news articles.

Although Random Forest and K-Nearest Neighbors achieved competitive performance, Logistic Regression consistently produced the highest accuracy while requiring lower computational resources and shorter training time. Multinomial Naïve Bayes also performed efficiently; however, its simplifying independence assumptions limited its classification performance compared to Logistic Regression.

VI. DISCUSSION

The experimental results demonstrate that Random Forest achieved the highest classification accuracy among the evaluated models. However, Logistic Regression was selected for deployment due to its competitive performance, lower computational complexity, and faster inference speed, making it more suitable for real-time fake news detection.

Although Random Forest achieved the highest accuracy, Logistic Regression was selected for deployment because it provides a better balance between predictive performance, computational efficiency, and inference speed. The use of TF-IDF feature extraction significantly improved the representation of textual data, enabling the classifier to learn meaningful linguistic patterns. These findings indicate that Logistic Regression is a suitable choice for real-time fake news detection applications.

VII. CONCLUSION

This paper presented a machine learning-based approach for fake news detection using Natural Language Processing (NLP) techniques. The proposed system employed text preprocessing and TF-IDF vectorization to transform textual news articles into numerical features for classification.

Four machine learning algorithms, namely Logistic Regression, Multinomial Naïve Bayes, Random Forest, and K-Nearest Neighbors, were trained and evaluated. Experimental results showed that Random Forest achieved the highest classification accuracy of 99.84%, while Logistic Regression demonstrated competitive performance with 98.96% accuracy and was selected as the deployment model due to its lower computational complexity, faster inference, and suitability for real-time applications.

The trained Logistic Regression model was successfully integrated into a Streamlit web application for real-time fake news prediction. The proposed system provides an efficient and reliable solution for automated fake news detection.

Future work may focus on incorporating transformer-based language models such as BERT, multilingual fake news detection, and multimodal analysis using textual and visual information.

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VIII. APPENDIX

The appendix contains representative Python code snippets used for text preprocessing, TF-IDF feature extraction, model training, and prediction. Sample test cases demonstrating fake and genuine news predictions are also included.

```
# -----  
# Apply Text Cleaning  
# -----  
  
print("\nCleaning text... Please wait.")  
  
news_data["content"] = news_data["content"].apply(clean_text)  
  
print("Text Cleaning Completed Successfully!")  
  
print("\nSample Cleaned Content:")  
print(news_data[["content", "label"]].head())
```

Fig. 3. Code Snippet.

```
=====
FAKE NEWS DETECTOR
=====

Enter News Article:

Aliens have signed a peace treaty with world leaders.

=====
Prediction : FAKE NEWS
Confidence : 81.19%
=====

Probability Distribution
Fake : 81.19%
Real : 18.81%
```

Fig. 4. Sample Prediction.